



Session 23

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Module 4, Week 11
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Module 4: **Core Supervised Learning**



Regression and Classification

Using

Sklearn (A tool kit/ML-package in python)

<https://colab.research.google.com/drive/1Nc8IL11q1SR3fwQMpaZ0OZPd6QakW1qD#scrollTo=1685a178>

What it is?



Major Machine Learning problems can be framed as:

1. Classification: Predict values for discrete (output) variable

Eg., Spam Classification (legitimate or spam email), Tumor Prediction (Benign or Malignant), Sentiment Classification (Positive or negative)

2. Regression: Predict values for a Continuous numerical (Output) variable

Eg., House Price Prediction, Temperature Forecasting, Solar energy forecasting, Stock price forecasting, etc.

Why these problems are Supervised?



- A significant amount of labeled (X and y) data available for supervision of ML algorithm.
- Input variables: one or more x (numeric or categorical).
- Output variable: y (discrete or continuous)

What is Linear Regression (LR)?



- Objective: Model the relationship between input variable and output variable (y) using labeled data and ML algorithm
- **Linear Regression** is a machine learning algorithm to model the relationship
- **x is independent**, while **y is dependent on x** .
- It is linear relation if y varies linearly with x , it can be quadratic, cubic, or more complicated relation.

Feature (X)

House Area
Years of Experience
Study Hours
Age

Target (Y)

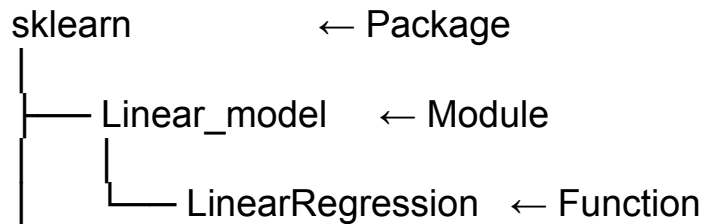
House Price
Salary
Exam Marks
Blood Pressure

Why Linear Regression?



- Fundamental ML algorithm to understand other algorithms
- Fast training as it assumes data is having linear relationship with y
- Highly interpretable due to simple formulation
- Good baseline model to compare with other ML algorithms
- Work perfectly for data having linear relationship, comparable to neural networks in case of linear relationship

How to implement LR in Python?



```
from sklearn.linear model import LinearRegression
```

- Sklearn Package (library) is The Scikit-learn machine learning package in python.
- Linear_model is a module inside sklearn containing linear machine learning models.
- LinearRegression is a class for creating Linear Regression model objects

Modeling



$y = mx + c$ // m is the slope (coefficient) of the line, c is the intercept (value of y at $x=0$)

$y = w_0 + w_1x$ // m and c are represented as weight (w) parameters in ML

Use existing labeled dataset to obtain optimal values of w parameters, then use the linear equation - $w_0 + w_1x$ to get predicted y for new/unseen x .

$Y' = 10 + 10x$ // optimal parameter values for the given data

Size (X)

100

200

300

400

500

Price (Y)

1010\$

2010\$

3010\$

4010\$

5010\$

\$ to keep values smaller

Modeling study hours for marks prediction

Interpretation: Base marks = 20

Every additional hour of study increases marks by approximately 8, so $8x$.

Optimal regression equation is $= 20 + 8x$

- Linear regression algorithm can learn the optimal values of weight parameters for the given historical data for any number of x variables.
- The model searches for the line that minimizes prediction errors.
- **Prediction Error: $\frac{1}{m} * \sum \text{over all } y ((y - y'))$** Mean Square Error (MSE)
// minimize sum of squared errors

Performance Evaluation



Three suitable metrics

1. Mean Absolute Error (MAE): Measures average absolute prediction error, lower is better.
2. Mean Square Error (MSE): Measure average square prediction error, lower is better // suitable for finding optimal parameter values also

Root Mean Squared Error (RMSE): Take root of the 2

3. R^2 Score: Measures goodness of fit between actual y and predicted y , higher is better, maximum value is 1.

Limitations of LR



- Will fail for data having complex relationship with y , other more sophisticated algorithms are the solution
- Outliers may spoil the model
- Assumes only one predictor x variable, in reality it can be more than one



Thank you!